

Quantitative Reasoning-I

Assignment 1 – Solution

BBA First Semester

Student Details (as per assignment)

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Business Scenario (Given Information)

A student starts a small coffee stall near a university. The following information is available:

- Monthly fixed cost = Rs. 30,000
- Cost of preparing one cup of coffee = Rs. 120
- Selling price of one cup = Rs. 250
- During one month, the stall sells 400 cups
- A customer survey shows that 80% of customers are satisfied
- Some customers comment: “The coffee is good, but the service is slow.”

Let: x = number of cups sold

Instructions

Answer all questions. Show calculations clearly and write short explanations in your own words. The assignment should not exceed two pages.

Q1. Quantitative and Qualitative Information (3 Marks)

a. Two examples of quantitative information from the case:

1. Monthly fixed cost = Rs. 30,000
2. Cost of preparing one cup = Rs. 120

(You could also mention: selling price Rs. 250, or 400 cups, or 80%.)

b. One example of qualitative information from the case:

- Some customers comment: “The coffee is good, but the service is slow.”

c. Difference between quantitative and qualitative information:

- **Quantitative information** is numerical and can be measured or counted (e.g., 30,000, 120, 250, 400, 80%).
- **Qualitative information** is descriptive and not numerical (e.g., “good coffee”, “service is slow”, “satisfied customers”).

Q2. Variables, Constants and Coefficients (3 Marks)

a. What does the variable x represent?

- x represents the number of cups sold.

b. Identify one constant from the case.

- One constant is 120 (cost of preparing one cup).
(Other constants: 30,000 or 250)

c. In the expression $120x$, what does 120 represent?

- 120 represents the cost per cup (the cost of preparing one cup).

Q3. Expressions and Basic Calculations (3 Marks)

- a. Write the expression for Total Cost (TC) when x cups are sold.

$$TC = \text{Fixed Cost} + (\text{Cost per cup} \times x)$$

$$TC = 30,000 + 120x$$

- b. Write the expression for Total Revenue (TR) when x cups are sold.

$$TR = \text{Selling Price per cup} \times x$$

$$TR = 250x$$

- c. If 400 cups are sold, calculate the Total Cost and Total Revenue.

$$TC = 30,000 + 120(400)$$

$$TC = 30,000 + 48,000 = 78,000$$

$$TR = 250(400)$$

$$TR = 100,000$$

Q4. Calculation and Interpretation (4 Marks)

- a. Calculate the Total Cost.

$$TC = 30,000 + 120(400)$$

$$TC = 30,000 + 48,000 = 78,000$$

- b. Calculate the Total Revenue.

$$TR = 250(400)$$

$$TR = 100,000$$

- c. Calculate the amount remaining after subtracting Total Cost from Total Revenue.

$$\text{Profit} = TR - TC$$

$$\text{Profit} = 100,000 - 78,000$$

$$\text{Profit} = 22,000$$

- d. Write one sentence explaining what the result means for the business.

The business makes a profit of Rs. 22,000 in a month after covering all the fixed and variable costs. This means the coffee stall is earning money and is financially viable (at 400 cups).

Quick Summary

- Total Cost (400 cups) = Rs. 78,000
- Total Revenue (400 cups) = Rs. 100,000
- Profit = Rs. 22,000
- 80% customers are satisfied (but service is slow).